TEK4090

Introduction to Modern Control

Lecture 6: Feedforward Control

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Back to Challenge 2:

- observer dynamics → $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$

We saw optimal gains for control, what about observers?

How to choose the observer gain optimally?

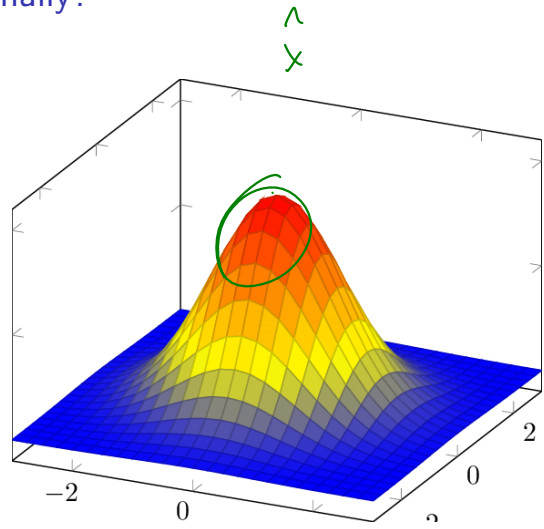
In general, measurements are **noisy**.

$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

$$E[v_k] = 0, \; E[w_k] = 0$$

$$E[v_k v_{k+1}^T] = \begin{cases} 0 & k \neq j \\ Q_k & k = j \end{cases}$$

$$E[w_k w_{k+1}^T] = \begin{cases} 0 & k \neq j \\ R_k & k = j \end{cases}$$

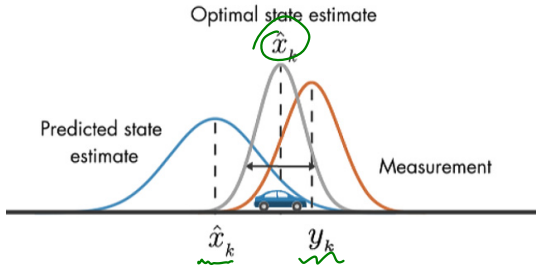


How to choose the observer gain?

With noisy measurements, the estimate is now a **random variable**.

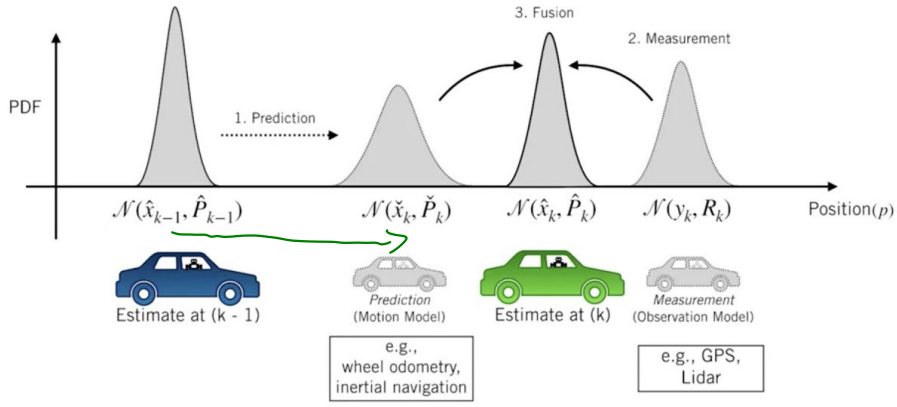
If L stabilizes the observer dynamics, then

$$E[e_k] = E[\hat{x}_k - x_k] \to 0.$$



Instead, we can optimize the 'width' (or, covariance) of the estimate

Optimal Observers



Kalman Filter Derivation

Truth model $x_{k+1} = Ax_k + Bu_k + Fw_k$ $\quad e_k = \hat{x}_k - x_k$

$\tilde{y}_k = Cx_k + v_k$

$E[v_k w_k^T] = 0$, $E[w_k w_j^T] = \begin{cases} 0 & k \neq j \\ Q_k & k = j \end{cases}$ $\quad E[v_k v_j^T] = \begin{cases} 0 & k \neq j \\ R_k & k = j \end{cases}$

at every timestep k, 2 things happen

propagate: $\hat{x}_{k+1}^- = A\hat{x}_k^+ + Bu_k$

update: $\hat{x}_k^+ = \hat{x}_k^- + K_k[\tilde{y}_k - C\hat{x}_k^-]$

+: updated w/ measurement

-: not updated

error covariance $P_k^\pm = E[e_k^\pm e_k^{\pm T}]$, $e_k^\pm = \hat{x}_k^\pm - x_k$

$e_{k+1}^- = Ae_k^+ - Fw_k$

Kalman Filter Derivation

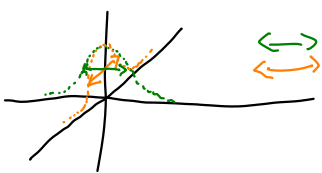
① $P_{k+1}^- = A P_k^+ A^T + F Q_k F^T$ → propagation of $P_k^+ \mapsto P_{k+1}^-$

For the update, we want P_k^+ in terms of K_k

② $\hat{x}_k^+ = (I - K_k C)\hat{x}_k^- + K_k C x_k + K_k v_k$

③ $e_k^+ = (I - K_k C) e_k^- + K_k v_k$

④ $P_k^+ = (I - K_k C) \underbrace{E[e_k^- e_k^{-T}]}_{P_k^-} (I - K_k C) + K_k R_k K_k$



$\Longleftrightarrow$ } widths are related to evals of P_k^+

↳ minimize trace of P_k^+

Kalman Filter Derivation

$\min_{K_k} \; Tr(P_k^+)$ s.t. eqn (4) 5 → take derivative with respect to P_k^+ (Matrix cookbook)

$$\frac{\partial}{\partial K_n} Tr(P_n^+) = -2(I - K_n C) P_n^- C^T + 2 K_n R_n = 0$$

$$K_n (C P_n^- C^T + R) = P_n^- C^T$$

$$\boxed{K_n = P_n^- C^T (C P_n^- C^T + R)^{-1}}$$

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter

Model: $x_{k+1} = Ax_k + Bu_k + Fv_k$

$y_k = Cx_k + Du_k + w_k$

Kalman Filter

Model: $x_{k+1} = Ax_k + Bu_k + Fv_k$
$y_k = Cx_k + Du_k + w_k$

Initialize: $\hat{x}_0 = x_0, \; P_0 = P$

Kalman Filter

Model: $x_{k+1} = Ax_k + Bu_k + Fv_k$
$y_k = Cx_k + Du_k + w_k$

Initialize: $\hat{x}_0 = x_0, \; P_0 = P$

Gain: $K_k = P_k^- C^T \left(CP_k^- C^T + R_k \right)^{-1}$

Kalman Filter

Model:
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Initialize:
$$\hat{x}_0 = x_0, \; P_0 = P$$

Gain:
$$K_k = P_k^- C^T \left(CP_k^- C^T + R_k \right)^{-1}$$

Update:
$$\hat{x}_k^+ = \hat{x}_k^- + K_k \left(\tilde{y}_k - C\hat{x}_k^- \right)$$
$$P_k^+ = \left(I - K_k C \right) P_k^-$$

Kalman Filter

Model:
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Initialize:
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Update:
$$\hat{x}_k^+ = \hat{x}_k^- + K_k \left(\tilde{y}_k - C\hat{x}_k^-\right)$$
$$P_k^+ = \left(I - K_k C\right) P_k^-$$

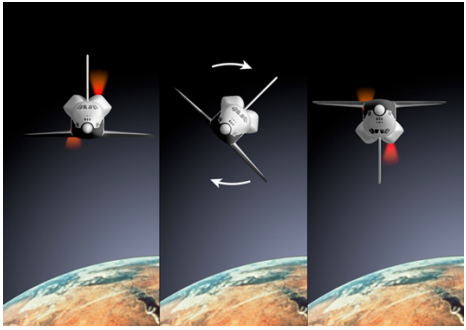
Propagate:
$$\hat{x}_{k+1}^- = A\hat{x}_k^+ + Bu_k$$
$$P_{k+1}^- = AP_k^+ A^T + BQ_k B^T$$

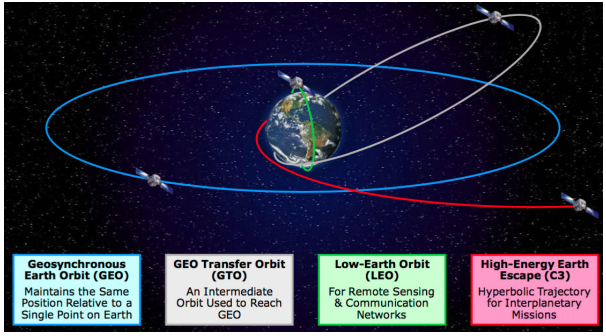
k+1 ↦ k

Guidance, Navigation, & Control

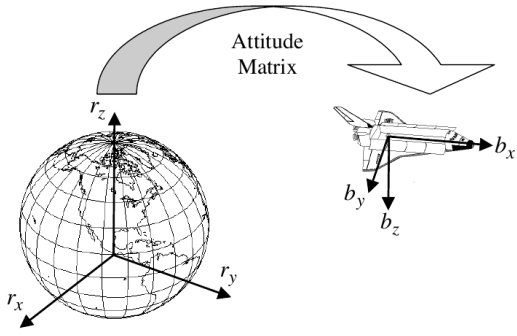
Spacecraft GNC

Spacecraft GNC





Attitude



Reference and body frames:

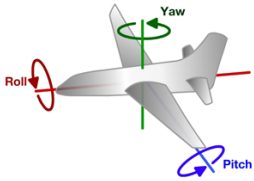
$$b = b_x\hat{b}_1 + b_y\hat{b}_2 + b_z\hat{b}_3$$
$$r = r_x\hat{r}_1 + r_y\hat{r}_2 + r_z\hat{r}_3$$

Transformation between the two:

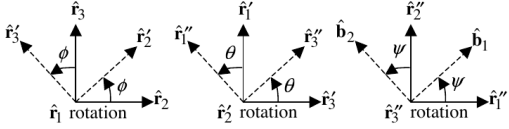
$$\begin{bmatrix}\hat{b}_1\\ \hat{b}_2\\ \hat{b}_3\end{bmatrix} = A\begin{bmatrix}\hat{r}_1\\ \hat{r}_2\\ \hat{r}_3\end{bmatrix} \quad (1)$$

Euler Angles

One can represent a rotation via the **Euler angles:** roll ϕ, pitch θ, and yaw ψ.

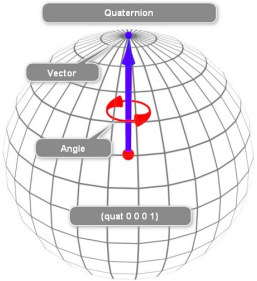


$$r' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} r \qquad r'' = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} r' \qquad b = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} r''$$



Quaternions

The Euler angles have a singularity when the attitude is pointed at a pole, so spacecraft attitude is represented as a **quaternion** instead.



$$q = \begin{bmatrix} \rho \\ q_4 \end{bmatrix} \in \mathbb{R}^4$$

$$q^T q = 1$$

$$A(q) = \Xi^T(q)\Psi(q)$$

$$\Xi(q) = \begin{bmatrix} q_4 I_{3\times 3} + [\rho\times] \\ -\rho^T \end{bmatrix}$$

$$\Psi(q) = \begin{bmatrix} q_4 I_{3\times 3} - [\rho\times] \\ -\rho^T \end{bmatrix}$$

$$[\rho\times] = \begin{bmatrix} 0 & -\rho_3 & \rho_2 \\ \rho_3 & 0 & -\rho_1 \\ -\rho_2 & \rho_1 & 0 \end{bmatrix}$$

matrix multiplication of $[\rho\times]$ is equivalent to cross product

Rigid Body Dynamics

Dynamics: Euler Angles

Kinematics: Quaternions

- H: Angular momentum

$$\dot{H} = J\dot{\omega}$$
$$= -[\omega\times]J\omega + L$$

$$\dot{q} = \frac{1}{2}\Omega(\omega)q$$
$$\Omega(\omega) = \begin{bmatrix} -[\omega\times] & \omega \\ -\omega^T & 0 \end{bmatrix}$$

The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

Rigid Body Dynamics

- H: Angular momentum
- J: Moment of Inertia

Dynamics: Euler Angles

$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega\times]J\omega + L\end{aligned}$$

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Rigid Body Dynamics

- H: Angular momentum
- J: Moment of Inertia
- L: Applied torque
- ω: Angular velocity

Dynamics: Euler Angles

$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega\times]J\omega + L\end{aligned}$$

Kinematics: Quaternions

$$\dot{q} = \frac{1}{2}\Omega(\omega)q$$

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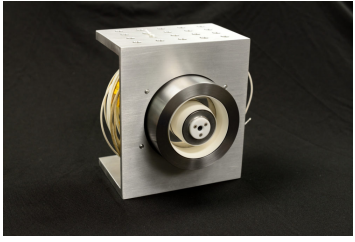
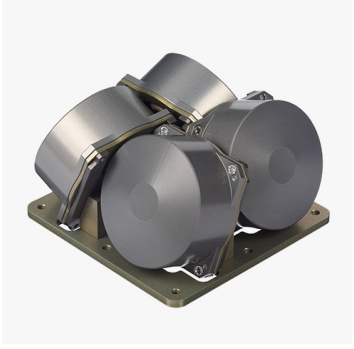
The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

Spacecraft Actuators

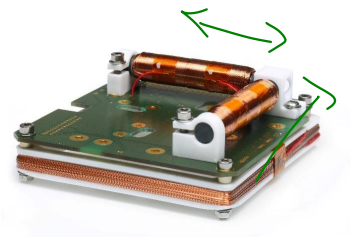
Thrusters



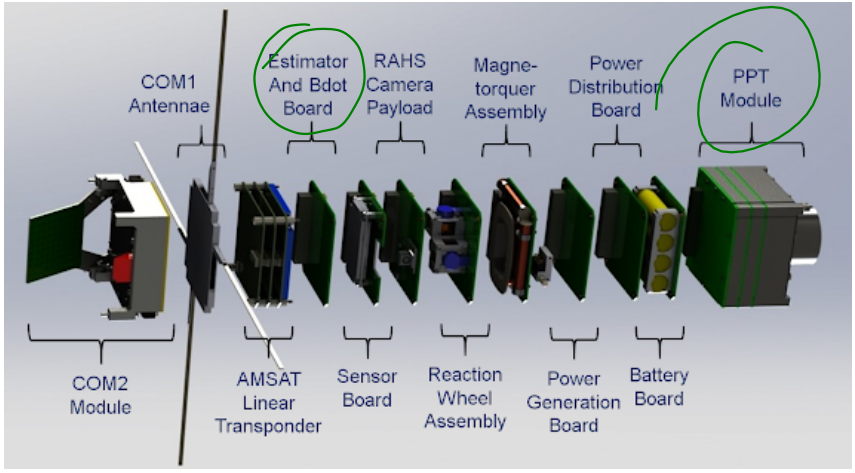
reaction wheels



Magnetorquers



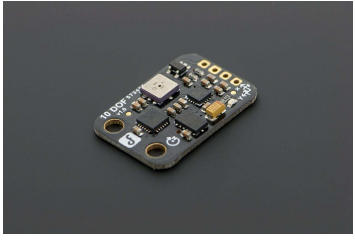
Plasma Thruster



Measurement

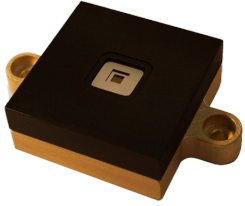
have a bias!

Inertial Measurement unit



Star tracker

Sun sensor

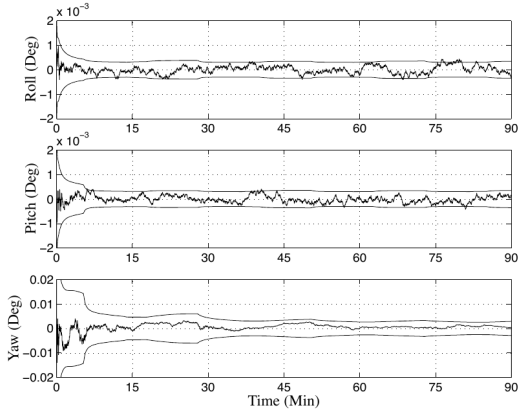


Quaternion Extended Kalman Filter

quatodynamics → nonlinear systems

Initialize	$\hat{\mathbf{q}}(t_0) = \hat{\mathbf{q}}_0, \quad \hat{\beta}(t_0) = \hat{\beta}_0$ $P(t_0) = P_0$
Gain	$K_k = P_k^- H_k^T(\hat{\mathbf{x}}_k^-)[H_k(\hat{\mathbf{x}}_k^-)P_k^- H_k^T(\hat{\mathbf{x}}_k^-) + R]^{-1}$ $H_k(\hat{\mathbf{x}}_k^-) = \begin{bmatrix} [A(\hat{\mathbf{q}}_k^-)\mathbf{r}_1\times] & 0_{3\times3} \\ \vdots & \vdots \\ [A(\hat{\mathbf{q}}_k^-)\mathbf{r}_n\times] & 0_{3\times3} \end{bmatrix}\Big\vert_{t_k}$
Update	$P_k^+ = [I - K_k H_k(\hat{\mathbf{x}}_k^-)]P_k^-$ $\Delta\hat{\mathbf{x}}_k^+ = K_k[\bar{\mathbf{y}}_k - \mathbf{h}_k(\hat{\mathbf{x}}_k^-)]$ $\Delta\hat{\mathbf{x}}_k^+ \equiv [\delta\hat{\alpha}_k^{+T} \; \Delta\hat{\beta}_k^{+T}]^T$ $\mathbf{h}_k(\hat{\mathbf{x}}_k^-) = \begin{bmatrix} A(\hat{\mathbf{q}}_k^-)\mathbf{r}_1 \\ A(\hat{\mathbf{q}}_k^-)\mathbf{r}_2 \\ \vdots \\ A(\hat{\mathbf{q}}_k^-)\mathbf{r}_n \end{bmatrix}\Big\vert_{t_k}$ $\hat{\mathbf{q}}_k^+ = \hat{\mathbf{q}}_k^- + \frac{1}{2}\Xi(\hat{\mathbf{q}}_k^-)\delta\hat{\alpha}_k^+, \quad$ re-normalize quaternion $\hat{\beta}_k^+ = \hat{\beta}_k^- + \Delta\hat{\beta}_k^+$
Propagation	$\hat{\omega}(t) = \omega(t) - \hat{\beta}(t)$ $\dot{\hat{\mathbf{q}}}(t) = \frac{1}{2}\Xi(\hat{\mathbf{q}}(t))\,\hat{\omega}(t)$ $\dot{P}(t) = F(t)P(t) + P(t)F^T(t) + G(t)Q(t)G^T(t)$ $F(t) = \begin{bmatrix} -[\hat{\omega}(t)\times] & -I_{3\times3} \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix}, \quad G(t) = \begin{bmatrix} -I_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & I_{3\times3} \end{bmatrix}$

bias



Matlab - simulating ODEs


```m
clear all; close all; clc;

% simulation timespan
dt=0.1;
t_space = [0:dt:10];

% initial condition
x_0 = [10;10];

% sinusoidal input with zero-order-hold
u = cos(t_space);
x_array = zeros(2,length(t_space));

x_array(:,1) = x_0;

for ii = 1:length(t_space)-1

    t_int = [t_space(ii), t_space(ii+1)];

    %simulate ODE with input
    [t_out,y_out] = ode45(@fn_rhs,t_int,x_array(:,ii),[],u(ii));
    x_array(:,ii+1) = y_out(end,:)';

end

 plot(t_space, x_array(1,:)), hold on
 plot(t_space, x_array(2,:))
```

ODE solvers

parameters for solver

pass u(ii) as a constant

initial cond

time domain

```m
function [x_dot] = fn_rhs(t, x, u)
%fn_rhs: a `right-hand-side function' of the ode x_dot = f(t,x,u)

x_dot = exp(-t)*[cos(x(1)); sin(x(2))] + exp(-0.1*t)*[0;1]*u;

end
```

Recap from Last Time

- Observability

Recap from Last Time

- Observability
- Estimators & State Estimation

Recap from Last Time

- Observability
- Estimators & State Estimation
- Optimal State Estimation: Kalman Filtering

Recap from Last Time

Note: all code available for Crassidis & Junkins

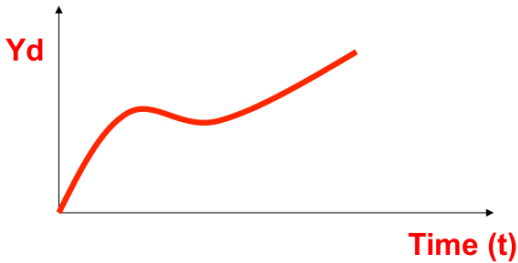
- Observability
- Estimators & State Estimation
- Optimal State Estimation: Kalman Filtering
- Applications to Spacecraft GNC

Reading Assignment

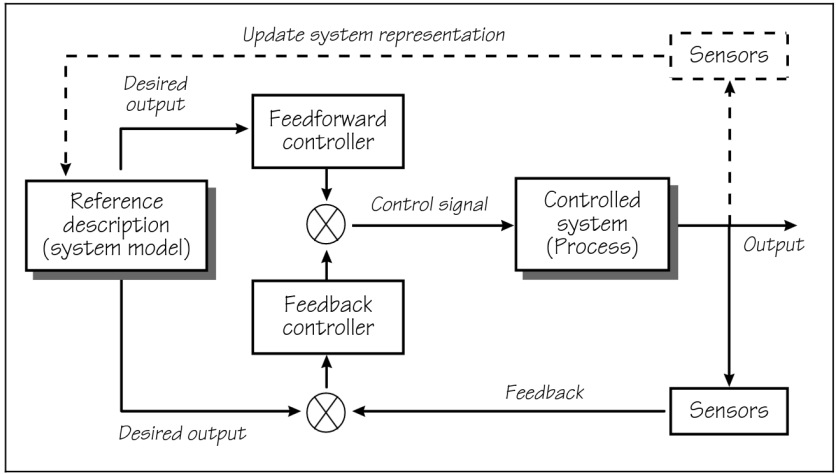
1. Notes by Santosh Devasia (University of Washington)

Motivation

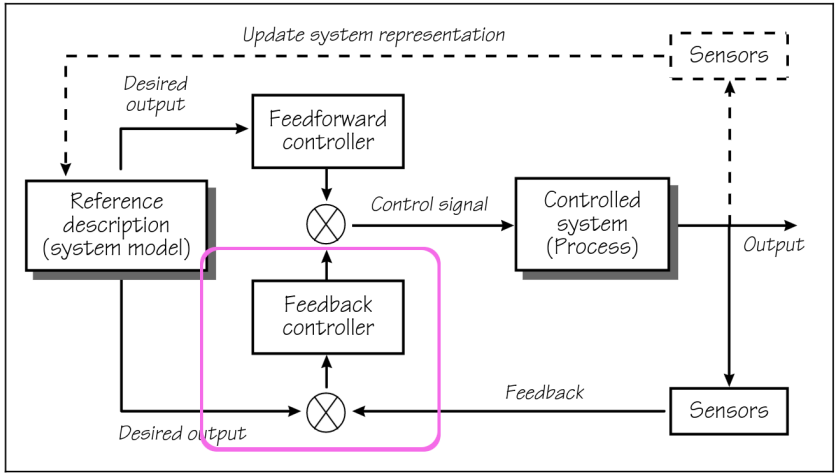
Goal: Find the input $u(t)$ that achieves a desired output trajectory $y(t)$



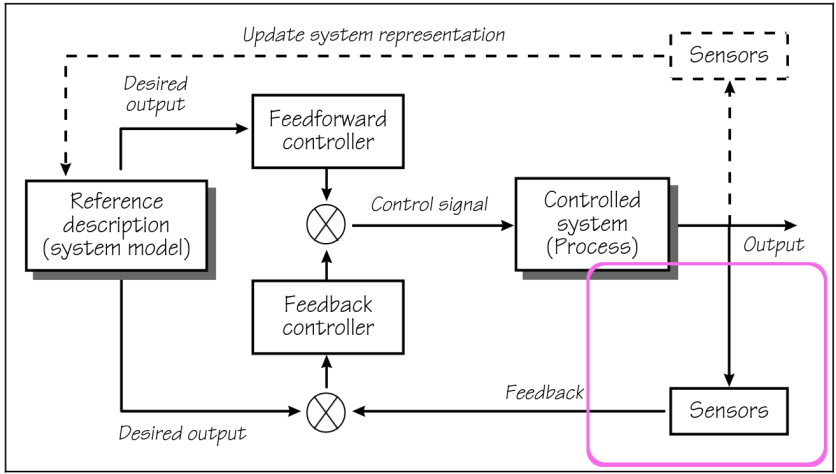
Control Block Diagram



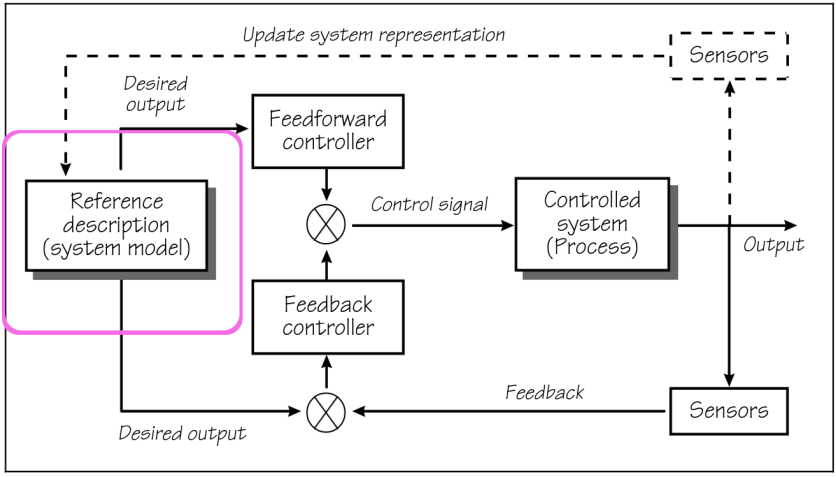
Control Block Diagram – Feedback



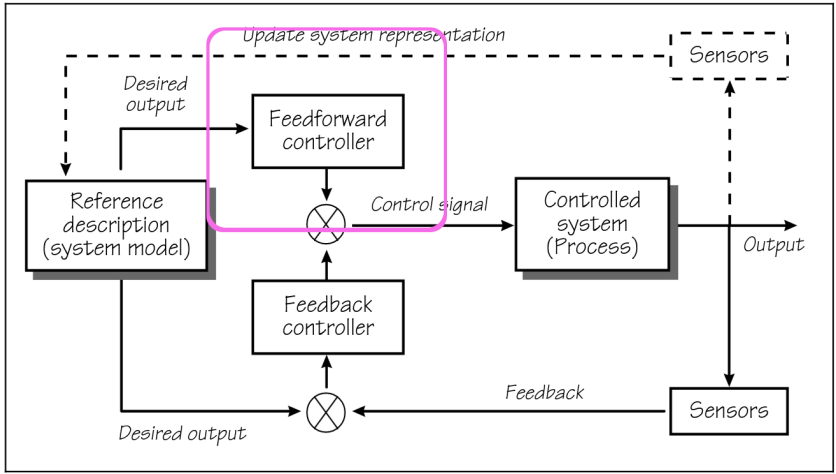
Control Block Diagram – Estimation



Control Block Diagram – System Identification



Control Block Diagram – Feedforward Control



System Inversion

$Y(s) = G(s)U(s)$ — calculate $Y(s)$, given $U(s)$

given $Y(s)$ $\{y(t)\}$ — how do we get $U(s)$ $\{u(t)\}$

Basic Example: Minimum-Phase SISO system

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s+1)}{(s+2)}$$

$$\implies Y(s) = \frac{(s+2)}{(s+1)}U(s)$$

System Inversion

Basic Example: Minimum-Phase SISO system

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s-1)}{(s-2)}$$

$$\implies Y(s) = \frac{(s-1)}{(s-2)} U(s)$$

Inverse Transfer Function

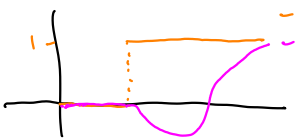
reciprocal

$$\frac{U(s)}{Y(s)} := G^{-1}(s) = \frac{(s-1)}{(s-2)}$$

$$\implies U(s) = \frac{(s-2)}{(s-1)} Y(s)$$

System Inversion

non-minimum phase: zeros of the TF (poles of inv TF) are positive



Basic Example: Minimum-Phase SISO system

Transfer Function

$$\frac{Y(s)}{U(s)} := G(s) = \frac{(s+1)}{(s+2)}$$
$$\implies Y(s) = \frac{(s+2)}{(s+1)}U(s)$$

Inverse Transfer Function

$$\frac{U(s)}{Y(s)} := G^{-1}(s) = \frac{(s+1)}{(s+2)}$$
$$\implies U(s) = \frac{(s+2)}{(s+1)}Y(s)$$

Note that both $G(s)$ and G^{-1} are **open-loop stable**

Setup

Linear SISO System

$$\dot{x} = Ax + Bu$$
$$y = Cx$$
$$u, y \in \mathbb{R}$$

Transfer Function

$$G(s) = C(sI - A)^{-1}B$$

Setup

"weaker" observability condition (controlability)

Assumption: Well-defined relative degree.

The system has a well-defined relative degree r if,

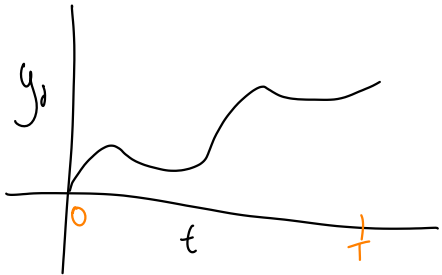
$$CA^j B = 0 \quad j < r-1$$
$$CA^{r-1}B \neq 0$$

scalar, $\Rightarrow$ invertible

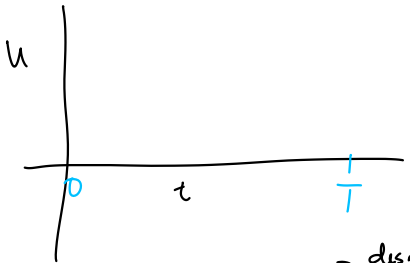
Setup

The relative degree r corresponds to the difference between the order of the denominator and the numerator of the transfer function

The Plan

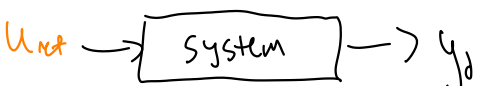


"inverse system"



$$\dot{x} = Ax + B u_{ref}$$

$$y = Cx = y_d(t)$$



$$\dot{\eta} = A_{inv}\eta + B_{inv} y_d$$

desired trajectory

$$u_{ref} = C_{inv}\eta + D_{inv} y_d$$

↳ output

Step 1: Differentiate Until the Input Appears

$$y = Cx$$

$$\dot{y} = C\dot{x} = C(Ax + Bu) = CAx + CBu$$

($CBu \to 0$)

$$\ddot{y} = CA\dot{x} = CA^2x + CABu$$

($CABu \to 0$)

$$\vdots$$

$$\frac{d^r}{dt^r}y = CA^rx + CA^{r-1}Bu(t)$$

($CA^{r-1}B \neq 0$ by assumption)

$$y^{(r)} = CA^rx + CA^{r-1}Bu$$

$CA^jB = 0$ if $j < r-1$

Step 1: Differentiate Until the Output Appears

$$y^{(r)} = \underbrace{CA^r x}_{A_y} + \underbrace{CA^{r-1}B}_{B_y}\overset{\neq 0}{u} = A_y x(t) + B_y u(t)$$

B_y → invertible

$$u_{inv} = B_y^{-1}\left(y^{(r)} - A_y x(t)\right)$$

problem: I don't know $x(t)$ (we know $y(t)$)

so, we can't easily get $x(t)$ from $y(t)$

Step 2: Decouple the State into Known/Unknown Parts

"known" parts

$$\begin{bmatrix} y \\ \dot{y} \\ \vdots \\ y^{(r-1)} \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{r-1} \end{bmatrix} x$$

full row rank, get x from y

$$z(t) = \begin{bmatrix} y_d \\ \dot{y}_d \\ \vdots \\ y^{(r-1)} \\ \hline \eta \end{bmatrix} := \begin{bmatrix} \eta_y \\ \hline \eta \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{r-1} \\ \hline T_\eta \end{bmatrix}}_{T} x = \begin{bmatrix} T_y \\ \hline T_\eta \end{bmatrix} x(t)$$

"unknown"

T_η is defined arbitrarily, so T is invertible

Step 2: Decouple the State into Known/Unknown Parts

$$x_{ref} = T^{-1} \begin{bmatrix} y_d \\ \eta_{ref} \end{bmatrix} \rightsquigarrow \eta(t),$$ so $x(t)$ does what it needs to do, so that $y = Cx = y_d$

$$= \underbrace{\left[T_\ell^{-1} \mid T_r^{-1} \right]}_{\text{notational abuse}} \begin{bmatrix} y_d \\ \eta_{ref} \end{bmatrix} = T_\ell^{-1} y_d + T_r^{-1} \eta_{ref}$$

next: plug into u_{inv} to get system of eqns for η_{ref}

Step 3: Write the Inverse Input Dynamics

$$u_{inv} = B_y^{-1}\left(y_d^{(r)} - A_y x_{ref}\right)$$

$$= B_y^{-1}\left(y_d^{(r)} - A_y\left(T_\ell^{-1} y_d + T_r^{-1}\eta_{ref}\right)\right)$$

$$= \underbrace{B_y^{-1} y_d^{(r)} - B_y^{-1} A_y T_\ell^{-1} y_d}_{\text{known}} - \underbrace{B_y^{-1} A_y T_r^{-1}\eta_{ref}}_{\text{unknown}}$$

$$= \underbrace{-B_y^{-1} A_y T_r^{-1}\eta_{ref}}_{} + \underbrace{\left[-B_y^{-1} A_y T_\ell^{-1} \mid B_y^{-1}\right]}_{D_{inv}} \underbrace{\begin{bmatrix} y_d \\ \hline y_d^{(r)} \end{bmatrix}}_{a_{y_d}}$$

Step 3: Write the Inverse Input Dynamics

$$u_{inv} = C_{inv}\eta(t) + D_{inv}y_d$$

$$z = \begin{bmatrix} y_d \\ \eta \end{bmatrix} = Tx \hookrightarrow \begin{bmatrix} T_y \\ T_\eta \end{bmatrix} x$$

Recover η dynamics from x

$\eta = T_\eta x$

short & fat matrix

Step 3: Write the Inverse Input Dynamics

$$\eta = T_\eta\, x$$

$$\dot\eta = T_\eta \dot x = T_\eta (Ax + Bu_{inv})$$

$$x_{ref} = T^{-1}\begin{bmatrix} y_d \\ \eta_{ref}\end{bmatrix} = \begin{bmatrix} T_\ell^{-1} \mid T_r^{-1}\end{bmatrix}\begin{bmatrix} y_d \\ \eta_{ref}\end{bmatrix} = T_\ell^{-1} y_d + T_r^{-1}\eta_{ref}$$

$$= T_\eta \left[Ax + B\left[-B_y^{-1}A_y T_r^{-1}\right]\eta + \left[-B_y^{-1}A_y T_\ell^{-1} \mid B_y^{-1}\right] y_d \right]$$

$$= T_\eta \left[A\left(T_\ell^{-1} y_d + T_r^{-1}\eta_{ref}\right) + B\left[-B_y^{-1}A_y T_r^{-1}\right]\eta + \left[-B_y^{-1}A_y T_\ell^{-1} \mid B_y^{-1}\right]\begin{bmatrix} y_d \\ \dot y_d \end{bmatrix}\right]$$

$$= T_\eta \left[AT_r^{-1} - BB_y^{-1}A_y T_r^{-1}\right]\eta + T_\eta \left[AT_\ell^{-1} - BB_y^{-1}A_y T_\ell^{-1} \mid BB_y^{-1}\right]\begin{bmatrix} y_d \\ y_d^{(r)}\end{bmatrix}$$

$$A_{inv} \qquad\qquad B_{inv}$$

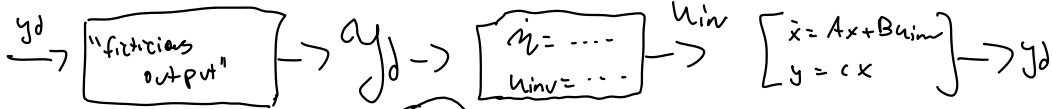
$$\boxed{\dot\eta = A_{inv}\eta + B_{inv}\, y_d}$$

Step 4: Write the Internal Dynamics

Step 4: Write the Internal Dynamics

Step 4: Write the Internal Dynamics

Step 5: Complete the Inverse System



$$A_{inv} = T_\eta \left[A - (BB_y^{-1}A_y) \right] T_r^{-1}$$

$$B_{inv} = \left[T_\eta \left[A - (BB_y^{-1}A_y) \right] T_l^{-1} \;\vdots\; T_\eta B B_y^{-1} \right]$$

$$C_{inv} = -B_y^{-1} A_y T_r^{-1}$$

$$D_{inv} = \left[-B_y^{-1} A_y T_l^{-1} \;\vdots\; B_y^{-1} \right]$$

$$\mathcal{Y}_d(t) = \begin{bmatrix} \xi_d(t) \\ y_d^{(r)}(t) \end{bmatrix}$$

$$\dot{\eta}_{ref} = A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d$$

$$u_{inv} = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$

A_{inv}

- eigenvalues of A_{inv} are the zeros of orginal system
- $\Rightarrow$ A_{inv} unstable for non-min phase systems

Step 5.5: Solve the Inverse System (min phase system)

(a) $\dot{\eta} = A_{inv}\eta + B_{inv}y_d$

$u_{inv} = C_{inv}\eta + D_{inv}y_d$

} ODE -> solve, need $\eta(t_0)$

-> assume the system starts at equilibrium, at t_0

$\dot{x} = 0 \Rightarrow \dot{\eta} = 0 = A_{inv}\eta(t_0) + B_{inv}y_d(t_0)$

$\eta(t_0) = -A_{inv}^{-1}B_{in}y_d(t_0)$

-> simulate (b)

Step 6: For non-minimum phase systems

Step 6: For non-minimum phase systems

Step 6: For non-minimum phase systems

Matlab Example

Non-Minimum Phase System

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \longrightarrow \begin{aligned} \dot{\eta}_{ref} &= A_{inv}\eta_{ref} + B_{inv}\mathcal{Y}_d \\ u_{inv} &= C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d \end{aligned} \longrightarrow \begin{aligned} \dot{\eta}_s &= A_s\eta_s + B_s\mathcal{Y}_d \\ \dot{\eta}_d &= A_u\eta_u + B_u\mathcal{Y}_d \end{aligned}$$

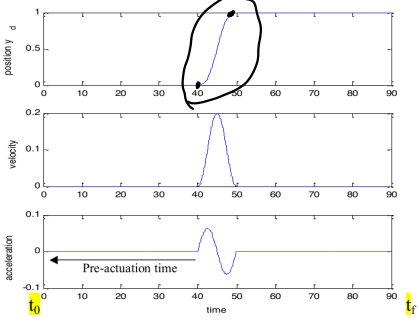
Focus: Internal Dynamics

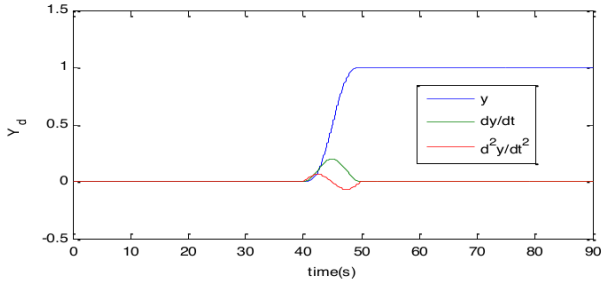
$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$
$$\dot{\eta}_d = A_u \eta_u + B_u \mathcal{Y}_d$$

$$A_s = -0.95$$
$$A_u = 1.05$$

Desired Trajectory

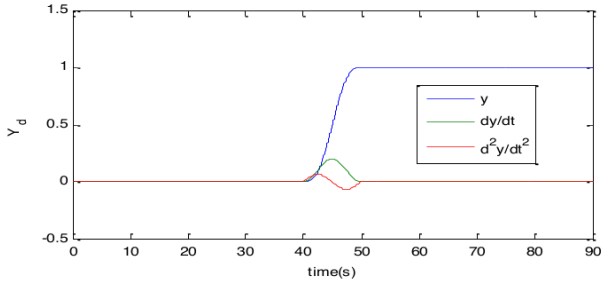
run it thru low pass filter





Solve Stable Subsystem

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$
$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$
$$C_s = \begin{bmatrix} 1 \end{bmatrix}, \; D_s = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Stable Subsystem

$$\dot{\eta}_s = A_s\eta_s + B_s\mathcal{Y}_d$$
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$$C_s = \begin{bmatrix}1\end{bmatrix},\ D_s = \begin{bmatrix}0 & 0 & 0\end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix}y & \dot{y} & \ddot{y}\end{bmatrix}^T$$

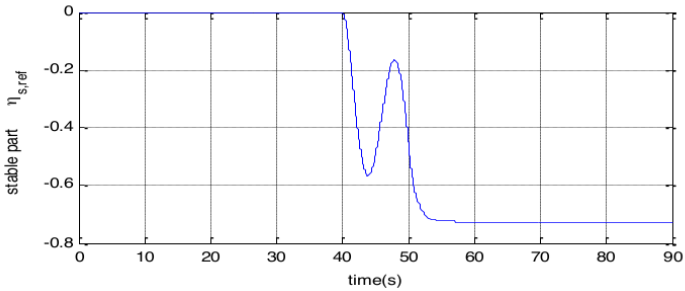
Initial condition (from equilibrium:)

$$0 = A_s\eta_s + B_s\mathcal{Y}_d$$
$$\eta_s(t_0) = -A_s^{-1}B_s\mathcal{Y}_d(t_0)$$

`sys_s=ss(`A_s`,` B_s`,` C_s`,` D_s`)`

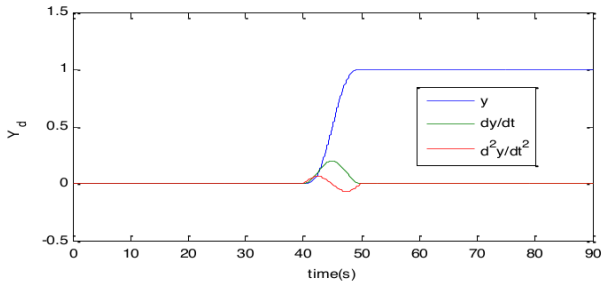
`[`$\mathcal{Y}_d$`, t,` $\eta_{s,ref}$`]=lsim(sys_s,` $\mathcal{Y}_d$`, t,` $\eta_s(t_0)$`)`

Solve Stable Subsystem



Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$
$$C_u = \begin{bmatrix} 1 \end{bmatrix}, \; D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$
$$C_u = \begin{bmatrix} 1 \end{bmatrix}, \; D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$

Solve it backwards in time:

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$
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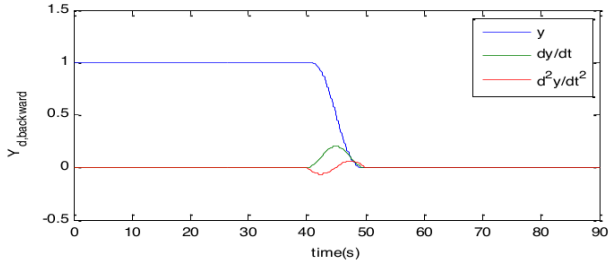


Solve Unstable Subsystem

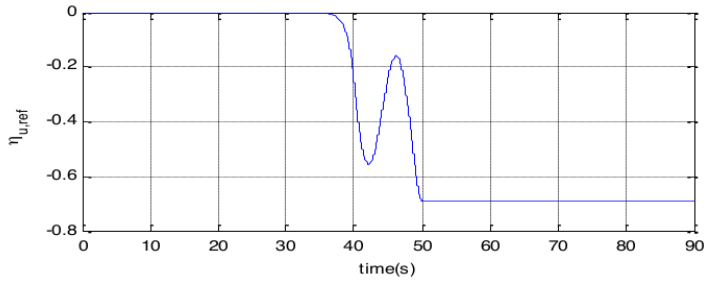
$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Initial condition (from equilibrium:)

$$0 = A_u \eta_u + B_u \mathcal{Y}_d$$
$$\eta_u(t_f) = -A_u^{-1} B_u \mathcal{Y}_d(t_f)$$

`sys_u_bw = ss(`$-A_u$`, `$-B_u$`, `C_u`, `D_u`)`
$\mathcal{Y}_{d,bw}$ `= fliplr(`$\mathcal{Y}_d$`)`
`[`$\mathcal{Y}_{ub}$`, t, `η_{ub}`] = lsim(sys_u_bw, `$\mathcal{Y}_{d,bw}$`, `$\eta_u(t_f)$`)`
$\eta_{u,ref}$ `= fliplr(`η_u^T`)`

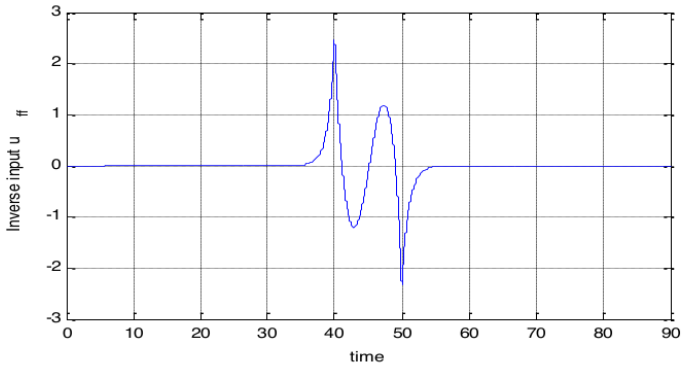
Solve Stable Subsystem



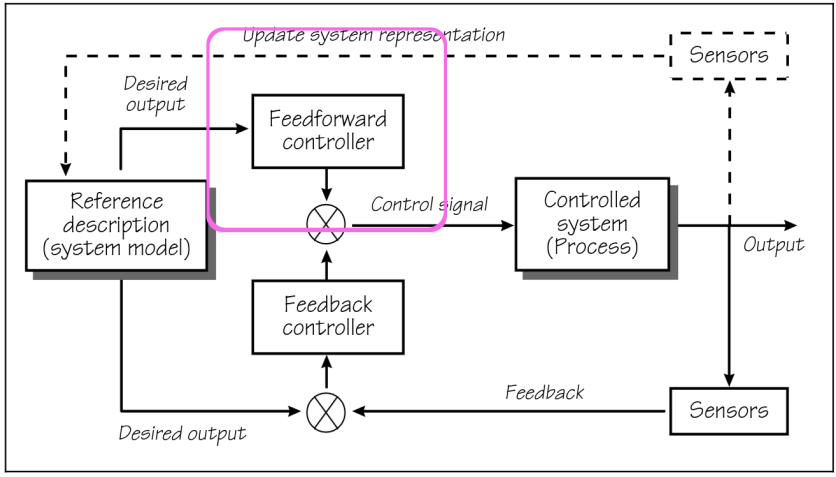
Find Reference Input

$$\eta_{ref} = T_{split}\begin{bmatrix}\eta_{s,ref}\\ \eta_{u,ref}\end{bmatrix}$$

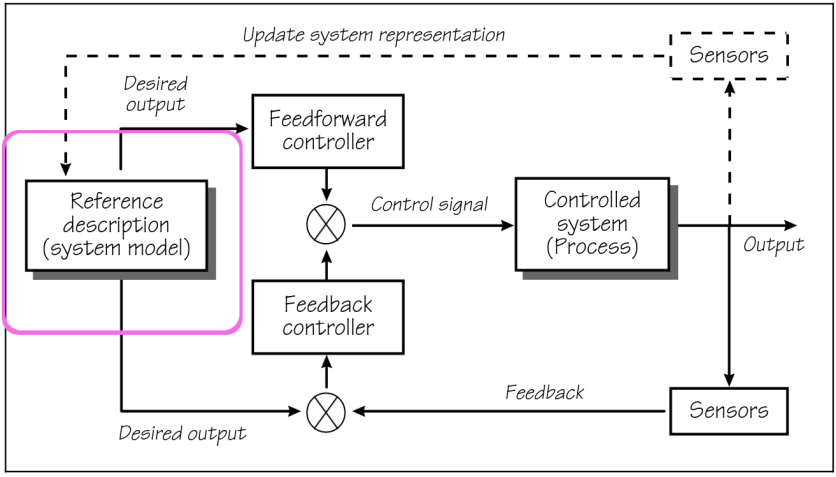
$$u_{inv}(t) = C_{inv}\eta_{ref} + D_{inv}\mathcal{Y}_d$$



Today


Update system representation
Sensors
Desired output
Feedforward controller
Reference description (system model)
Control signal
Controlled system (Process)
Output
Feedback controller
Feedback
Sensors
Desired output

Next Time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